The theory of free Steiner triple systems is stable

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Editorial note. The counted statement is the author's environment at source0.tex lines 1033-1034 and its proof environment opens at line 1035; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

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Editorial scope. One original result of the pinned author manuscript, counted once: stability of the theory of free Steiner triple systems (author label stable).

Allowed standard prerequisites (editorial).

- Stability. A theory is stable when over every model the number of types in one variable is at most the cardinality of the model raised to the power omega.. It is the conclusion of the statement.
- Type over a model. A type is a consistent set of formulas in one free variable with parameters in the model.. It is what the proof counts.
- Isomorphism type of a countable tuple. Countable tuples over a model are classified up to partial isomorphism by their quantifier-free type in the language of the partial structure.. It is the invariant that bounds the number of types.

Corollary 0.1 If $M_1 \preccurlyeq M_2$ are unconfined STSs, then $M_1 \leq M_2$.

Lemma 0.2 Let M be an infinite unconfined STS and let $\bar{a}, \bar{b}$ be tuples from M. If for some HF-orderings $<$ and $<'$ of M, $\mathrm{cl}_<(\bar{a})$ and $\mathrm{cl}_{<'}(\bar{b})$ have the same diagram as partial STS, then $\mathrm{tp}_M(\bar{a}) = \mathrm{tp}_M(\bar{b})$.

Proposition 0.3 The theory of free STSs is stable.

Proof. We count 1-types over a model M and show that their number is $\leq |M|^\omega$. We assume M is an elementary substructure of the monster model $\mathbb{M}$. Then $M \leq \mathbb{M}$ by Corollary 0.1, and we can fix an HF-ordering $<$ of $\mathbb{M}$ where M is an initial segment. If $a, b \in \mathbb{M}$ and $\mathrm{cl}_<(a)$, $\mathrm{cl}_<(b)$ have the same diagram over M as partial STS, then, by Lemma 0.2, they have same type over M. Hence, the number of 1-types over M is bounded by the number of partial STS isomorphism types of countable tuples over M, which is clearly $\leq |M|^\omega$. □